

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 09

DATE : 07-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 09

Topics

DATE : 07 - 12 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : S.H.M. Elasticity, Waves on String, Sound Waves

CHEMISTRY : Alcohol + Ether, Group-13 and 14

ZOOLOGY : Respiratory System, Human Health & Disease Including AIDS & Cancer

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

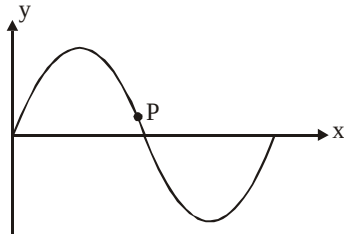
- | | |
|---|---|
| <p>1. A simple pendulum performs simple harmonic motion about $X = 0$ with an amplitude A and time period T. The speed of the pendulum at $X = \frac{A}{2}$ will be</p> <p>(1) $\frac{\pi A \sqrt{3}}{T}$ (2) $\frac{\pi A}{T}$</p> <p>(3) $\frac{\pi A \sqrt{3}}{2T}$ (4) $\frac{3\pi^2 A}{T}$</p> <p>2. A body of mass 5 gm is executing S.H.M. about a point with amplitude 10 cm. Its maximum velocity is 100 cm/sec. Its velocity will be 50 cm/sec at a distance</p> <p>(1) 5 (2) $5\sqrt{2}$</p> <p>(3) $5\sqrt{3}$ (4) $10\sqrt{2}$</p> <p>3. A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is</p> <p>(1) 2π sec (2) $\pi/2$ sec</p> <p>(3) π sec (4) $3\pi / 2$ sec</p> <p>4. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is</p> <p>(1) $\sqrt{3}$ cm (2) $\sqrt{5}$ cm</p> <p>(3) $2(\sqrt{3})$ cm (4) $2(\sqrt{5})$ cm</p> | <p>5. The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is 16 cm/sec. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}$ cm/s, will be</p> <p>(1) $2\sqrt{3}$ cm (2) $\sqrt{3}$ cm</p> <p>(3) 1 cm (4) 2 cm</p> <p>6. The displacement equation of a particle is $x = 3\sin 2t + 4\cos 2t$. The amplitude and maximum velocity will be respectively</p> <p>(1) 5, 10 (2) 3, 2</p> <p>(3) 4, 2 (4) 3, 4</p> <p>7. The instantaneous displacement of a simple pendulum oscillator is given by $x = A \cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time :-</p> <p>(1) $\frac{\pi}{4\omega}$ (2) $\frac{\pi}{2\omega}$</p> <p>(3) $\frac{\pi}{\omega}$ (4) $\frac{2\pi}{\omega}$</p> <p>8. A small body of mass 0.10 kg is executing S.H.M. of amplitude 1.0 m and period 0.20 sec. The maximum force acting on it is</p> <p>(1) 100 N (2) 120 N</p> <p>(3) 102 N (4) 80 N</p> |
|---|---|

SPACE FOR ROUGH WORK

9. A body executing simple harmonic motion has a maximum acceleration equal to 24 metres/sec² and maximum velocity equal to 16 metres/sec. The amplitude of the simple harmonic motion is
- (1) $\frac{32}{3}$ metres (2) $\frac{3}{32}$ metres
- (3) $\frac{1024}{9}$ metres (4) $\frac{64}{9}$ metres
10. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is
- (1) $\frac{1}{2\pi\sqrt{3}}$ (2) $2\pi\sqrt{3}$
- (3) $\frac{2\pi}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2\pi}$
11. A particle is vibrating in a simple harmonic motion with an amplitude of 4 cm. At what displacement from the equilibrium position, is its energy half potential and half kinetic (Potential energy at mean position is zero)
- (1) 1 cm (2) $\sqrt{2}$ cm
- (3) 3 cm (4) $2\sqrt{2}$ cm
12. The potential energy of a particle executing S.H.M. is 2.5 J, when its displacement is half of amplitude. The total energy of the particle be (Potential energy at mean position is zero)
- (1) 18 J (2) 10 J
- (3) 12 J (4) 2.5 J
13. The angular velocity and the amplitude of a simple pendulum is ω and a respectively. At a displacement X from the mean position if its kinetic energy is T and potential energy is V , then the ratio of T to V is (Potential energy at mean position is zero)
- (1) $X^2\omega^2/(a^2 - X^2\omega^2)$ (2) $X^2/(a^2 - X^2)$
- (3) $(a^2 - X^2\omega^2)$ (4) $(a^2 - X^2)/X^2$
14. The total energy of a particle executing S.H.M. is 80 J. What is the potential energy when the particle is at a distance of $3/4$ of amplitude from the mean position
- (1) 60 J (2) 10 J
- (3) 40 J (4) 45 J
15. A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as a function of displacement x . Which of the following statements is true
- (1) P.E. is maximum when $x = 0$
- (2) K.E. is maximum when $x = 0$
- (3) T.E. is zero when $x = 0$
- (4) K.E. is maximum when x is maximum
16. The total energy of a particle, executing simple harmonic motion is
- (1) $\propto x$ (2) $\propto x^2$
- (3) Independent of x (4) $\propto x^{1/2}$
17. The equation of motion of a particle is $\frac{d^2y}{dt^2} + Ky = 0$, where K is positive constant. The time period of the motion is given by
- (1) $\frac{2\pi}{K}$ (2) $2\pi K$
- (3) $\frac{2\pi}{\sqrt{K}}$ (4) $2\pi\sqrt{K}$

SPACE FOR ROUGH WORK

18. A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. The wavelength of the wave is 0.5m and its amplitude is 10 cm. At a particular time t , the snap-shot of the wave is shown in fig. The velocity of P when its displacement is 5 cm is :-



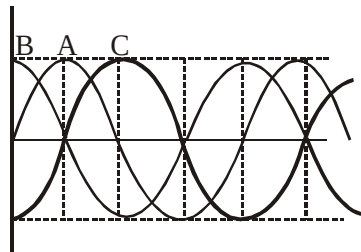
- (1) $\frac{\sqrt{3}\pi}{50} \hat{i}$ m/s (2) $-\frac{\sqrt{3}\pi}{50} \hat{j}$ m/s
(3) $\frac{\sqrt{3}\pi}{50} \hat{j}$ m/s (4) $-\frac{\sqrt{3}\pi}{50} \hat{i}$ m/s
19. The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is:-
(1) 44% (2) 40%
(3) 20% (4) 10%
20. A particle describes SHM in a straight line about O.



If the time period of the motion is T then its kinetic energy at P be half of its peak value at O, if the time taken by the particle to travel from O to P is

- (1) $\frac{1}{2}T$ (2) $\frac{1}{4}T$ (3) $\frac{1}{2\sqrt{2}}T$ (4) $\frac{1}{8}T$

21. The speed of transverse waves in a stretched string is 700 cm/s. If the string is 2 m long, the frequency with which is resonates in fundamental mode is :-
(1) $(7/2)$ Hz (2) $(7/4)$ Hz
(3) (14) Hz (4) $(2/7)$ Hz
22. A body executes S.H.M. with an amplitude A . At what displacement from the mean position, is the potential energy of the body one-fourth of its total energy?
(1) $\frac{A}{4}$ (2) $\frac{A}{2}$
(3) $\frac{3A}{4}$
(4) Some other fraction of A
23. The time period of oscillations of a simple pendulum is 1 minute. If its length is increased by 44%, then its new time period of oscillation will be:-
(1) 96 s (2) 58 s (3) 82 s (4) 72 s
24. Three progressive wave A, B and C are shown in fig. with respect to wave A :-



- (1) the wave C lags behind in phase by $\pi/2$ and B leads by $\pi/2$
(2) the wave C leads in phase by π and B lags behind by π
(3) the wave C leads in phase by $\pi/2$ and B lags behind by $\pi/2$
(4) the wave C lags behind in phase by π and B leads by π

SPACE FOR ROUGH WORK

25. The equation of motion of a particle executing simple harmonic motion is $a + 16\pi^2x = 0$. In this equation, a is the linear acceleration in m/s^2 of the particle at a displacement x in metre. Find the time period.

(1) 0.50 (2) 0.15 (3) 0.155 (4) 0.25

26. A wave equation which gives the displacement along y -direction is given by; $y = 10^{-4} \sin(60t + 2x)$

where x and y are in m and t in s . This represents a wave :-

(i) travelling with velocity of 30 m/s in the negative x -direction

(ii) of wavelength $(\pi) \text{ m}$

(iii) of frequency $(30/\pi) \text{ Hz}$

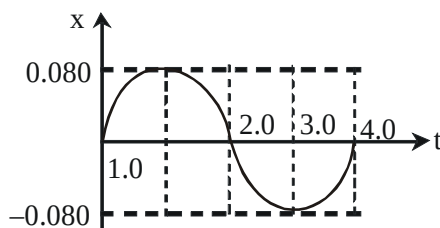
(iv) of amplitude 10^{-4} m travelling along negative x -axis

(1) (iii, iv) (2) (ii, iv)
(3) (ii, iii) (4) (i, ii, iii, iv)

27. A particle of mass m is executing S.H.M. If amplitude is a and frequency n , the value of its force constant will be :

(1) mn^2 (2) $4mn^2a^2$
(3) ma^2 (4) $4\pi^2mn^2$

28. An object of mass 0.8 kg is attached to one end of a spring and the system is set into simple harmonic motion. The displacement x of the object as a function of time is shown in the figure. The magnitude of the object's acceleration at $t = 1.0$ is



(1) zero
(2) 1.57 m/s^2
(3) 0.197 m/s^2
(4) 0.157 m/s^2

29. A particle executes SHM of type $x = a \sin \omega t$. It takes time t_1 from $x = 0$ to

$x = \frac{a}{2}$ and t_2 from $x = \frac{a}{2}$ to $x = a$. The ratio of $t_1 : t_2$ will be :-

(1) 1 : 1 (2) 1 : 2
(3) 1 : 3 (4) 2 : 1

30. The diagram below shows an instantaneous position of a string as a transverse progressive wave travels along it from left to right



Which one of the following correctly shows the direction of the velocity of the points 1, 2 and 3 on the string ?

1	2	3
(1) $\rightarrow$	$\rightarrow$	$\rightarrow$
(2) $\rightarrow$	$\leftarrow$	$\rightarrow$
(3) $\downarrow$	$\downarrow$	$\downarrow$
(4) $\downarrow$	$\uparrow$	$\downarrow$

31. A transverse wave on a string with a linear mass density of 0.20 kg/m is described by the following equation $y = 0.05 \sin(420t - 21.0x)$

where x and y are in metres and t is seconds. The tension in the string is equal to ?

(1) 32 N (2) 42 N
(3) 66 N (4) 80 N

SPACE FOR ROUGH WORK

- | | |
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| <p>32. In a stationary wave all the particles :-
 (1) On either side of a node vibrate in same phase
 (2) In the region between two nodes vibrate in same phase
 (3) In the region between two antinodes vibrate in same phase
 (4) Of the medium vibrate in same phase</p> <p>33. The particle displacement (in cm) in a stationary wave is given by $y(x, t) = 2 \sin(0.1 \pi x) \cos(100 \pi t)$. The distance between a node and the next antinode is :-
 (1) 2.5 cm (2) 7.5 cm
 (3) 5 cm (4) 10 cm</p> <p>34. Two wires are kept tight between the same pair of support. The tensions in the wires are in the ratio 2 : 1, the radii are in the ratio 3 : 1 and the densities are in the ratio 1 : 2. The ratio of their fundamental frequencies is :-
 (1) 2 : 3 (2) 2 : 4
 (3) 2 : 5 (4) 2 : 6</p> <p>35. A wave is reflected from a rigid support. The change in phase on reflection will be
 (1) $\pi/4$ (2) $\pi/2$
 (3) π (4) 2π</p> <p>36. There is a destructive interference between the two waves of wavelength λ coming from two different paths at a point. To get maximum sound or constructive interference at that point, the path of one wave is to be increased by
 (1) $\frac{\lambda}{4}$ (2) $\frac{\lambda}{2}$
 (3) $\frac{3\lambda}{4}$ (4) λ</p> | <p>37. When a wave passes from one medium to another, there is change in
 (1) frequency and velocity
 (2) frequency and wavelength
 (3) wavelength and velocity
 (4) frequency, wavelength and velocity</p> <p>38. When an ultrasonic wave travels from air into water
 (1) the wave bends away from the normal
 (2) the wave bends towards the normal
 (3) wave goes undeviated
 (4) concept of bending of wave is only applicable to light waves</p> <p>39. Transverse mechanical wave are not possible in liquid. Because :
 (1) liquid having bulk modulus
 (2) liquid volume is constant
 (3) No shearing stress in liquid
 (4) liquid density is constant</p> <p>40. Two mechanical waves,
 $y_1 = 2 \sin 2\pi(50 t - 2x)$ &
 $y_2 = 4 \sin 2\pi(ax + 100 t)$ propagate in a medium with same speed.
 (1) The ratio of their intensities is 1: 16
 (2) The ratio of their intensities is 1: 4
 (3) The value of 'a' is 8 units
 (4) The value of 'a' is 2 units</p> |
|--|---|

SPACE FOR ROUGH WORK

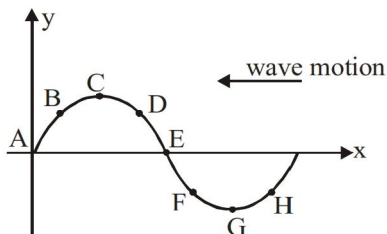
41. A transverse harmonic wave on a string

is described by $y = 3\sin[36t + 0.018x + \frac{\pi}{4}]$

where x and y are in cm and t in s. The least distance between two successive crests in the wave is :

- (1) 2.5 m (2) 3.5 m
(3) 1.5 m (4) 4.5 m

42. A transverse wave is travelling along a stretched string from right to left. The figure shown represents the shape of the string (snap short) at a given instant. At this instant

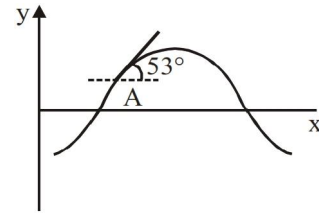


- (1) the particles at A, B and H have downward velocity
(2) the particles at D, E and F have downward velocity
(3) the particles at C, E and G have zero velocity
(4) the particles at A and F have maximum velocity

43. A transverse wave in a medium is described by the equation $y = A \sin^2(\omega t - kx)$. The magnitude of the maximum velocity of particles in the medium is equal to that of the wave velocity, if the value of A is $\{\lambda = \text{wavelength of wave}\}$:-

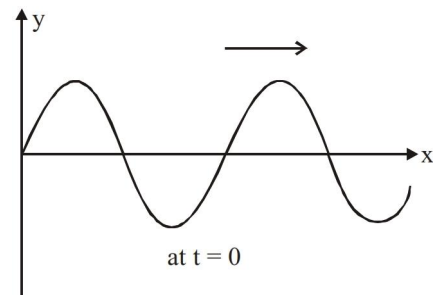
- (1) $\lambda / 2\pi$ (2) $\lambda / 4\pi$
(3) λ / π (4) $2\lambda / \pi$

44. Curve 'y-x' at an instant for a wave travelling along x-axis on a string is shown. Slope at the point A on the curve, as shown, is 53° .



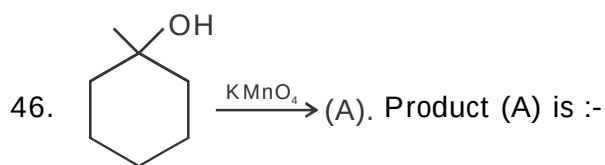
- (1) Transverse velocity of the particle at point A is positive if the wave is travelling along positive x-axis.
(2) Transverse velocity of the particle at point A is positive if the wave is travelling along negative x-axis
(3) Magnitude of transverse velocity of the particle at point A is lesser than wave speed.
(4) Both (1) & (3)

45. Figure shows a sinusoidal plot which describe the wave. Select the correct equation :-

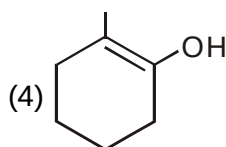
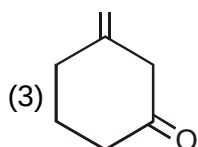
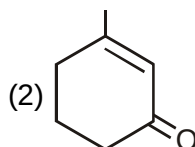


- (1) $y = A \sin(\omega t - kx)$
(2) $y = A \sin(kx - \omega t)$
(3) $y = A \sin(\omega t + kx)$
(4) $y = -A \sin(\omega t + kx)$

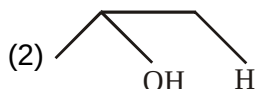
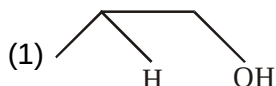
SPACE FOR ROUGH WORK



(1) No reaction



Product 'A' is :-



(3) Both (1) & (2)

(4) Ethene

48. Secondary alcohol on victor mayer test gives :-

(1) Red colour

(2) Colourless solution

(3) Blue colour

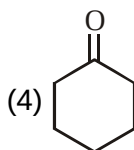
(4) Pink colour

49. A compound (X) on reaction with PCl_5 produced compound (Y), POCl_3 and a gas (Z). When this gas comes to contact with NH_3 , white fumes are observed, compound (X) can be :-

(1) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$

(2) $\text{CH}_3 - \text{CHO}$

(3) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$



50. 2-phenyl propene on acidic hydration gives

(1) 2-phenyl-2-propanol

(2) 2-phenyl-1-propanol

(3) 3-phenyl-1-propanol

(4) 1-phenyl-2-propanol

51. OsO_4 is used for

(1) Hydroxylation of acetylenes

(2) Hydroxylation of olefins to form cis-diol

(3) Hydroxylation of olefine to form trans-diol

(4) Hydroxylation of carbonyl compounds

52. Alcohol (X) $\xrightarrow{\text{aq. NaOH} + \text{I}_2}$ CHI_3 + (Y)

$\xrightarrow{\text{H}_3\text{O}^+}$ PhCH_2COOH . The alcohol(X) is

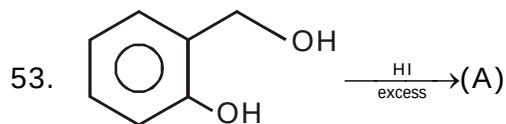
(1) $\text{CH}_3\underset{\text{Ph}}{\text{CH}}\text{CH}(\text{OH})\text{CH}_3$

(2) $\text{PhCH}_2\text{CH}(\text{OH})\text{CH}_3$

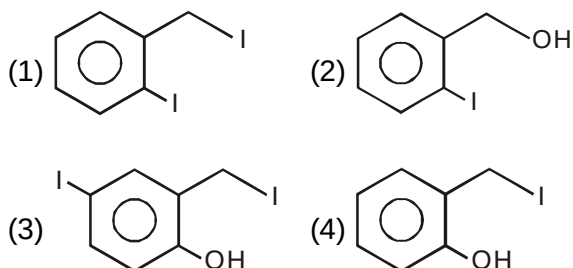
(3) $\text{PhCH}(\text{OH})\text{CH}_2\text{CH}_3$

(4) $\text{PhCH}(\text{CH}_3)\text{OH}$

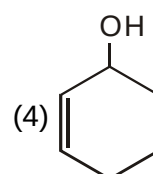
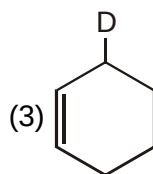
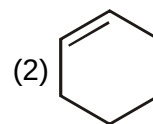
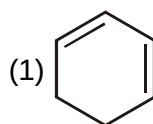
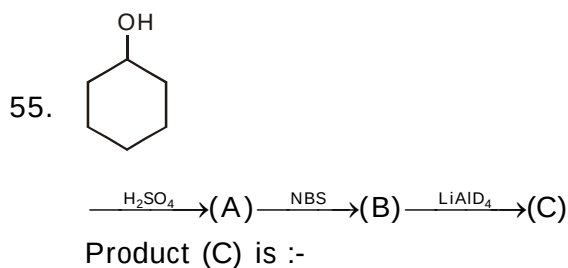
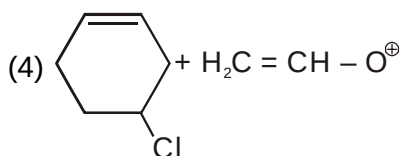
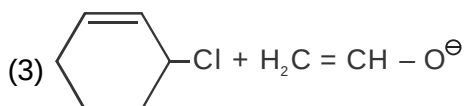
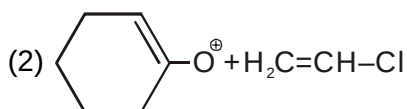
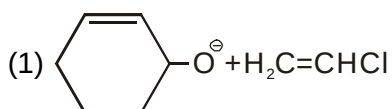
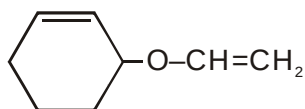
SPACE FOR ROUGH WORK



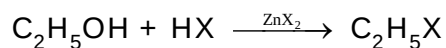
Major Product (A) is :-



54. The best method for synthesis of given ether by Williamson's ether synthesis is :-



56. For the reaction,



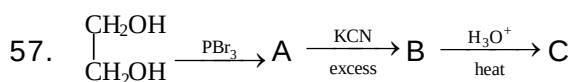
The decreasing order of reactivity of halogen acids is

(1) $\text{HI} > \text{HCl} > \text{HBr}$

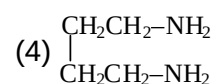
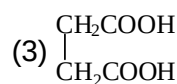
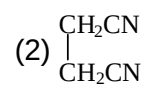
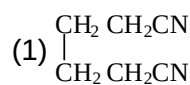
(2) $\text{HI} > \text{HBr} > \text{HCl}$

(3) $\text{HCl} > \text{HBr} > \text{HI}$

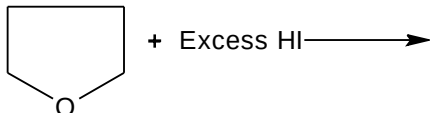
(4) $\text{HBr} > \text{HI} > \text{HCl}$



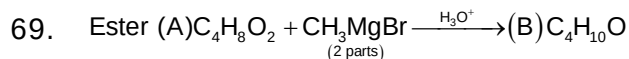
In the above sequence of reactions the end product (C) is



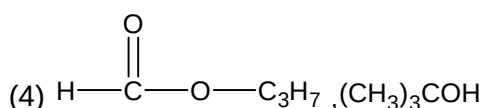
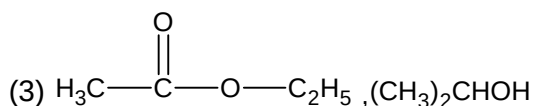
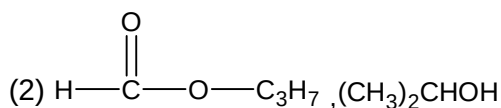
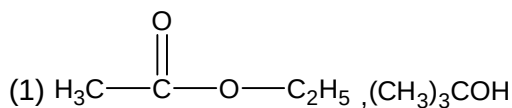
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58. The product of reaction $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[\text{Cu}]{300^\circ\text{C}} ?$ is :
 (1) C_2H_6 (2) CH_3COCH_3
 (3) CH_3COOH (4) CH_3CHO
59. In the following sequence of reactions,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{P}+\text{I}_2} \text{A} \xrightarrow[\text{ether}]{\text{Mg}} \text{B}$
 $\xrightarrow{\text{HCHO}} \text{C} \xrightarrow{\text{H}_2\text{O}} \text{D}$, then compound 'D' is
 (1) butanal
 (2) n-butyl alcohol
 (3) n-propyl alcohol
 (4) propanal
60. Which of the following is a tertiary alcohol ?
 (1) 2-Methylpropan-1-ol
 (2) 2-Methylpropan-2-ol
 (3) 3-Methylbutan-1-ol
 (4) 3-Methylbutan-2-ol
61. $\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{OH} \xrightarrow[-\text{H}_2\text{O}]{\text{H}^+} [\text{F}] \xrightarrow[\text{CCl}_4]{\text{Br}_2} \text{C}_4\text{H}_8\text{Br}_2$
 5 such products are possible
 How many isomers of [F] are possible ?
 (1) 2 (2) 5 (3) 6 (4) 3
62. The reaction of $\text{CH}_3\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives :
 (1) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{OH}$
 (2) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{OH}$
 (3) $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{Br}$
 (4) $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{Br}$
63. In the Lucas test of alcohols, appearance of cloudiness is due to the formation of
 (1) aldehyde (2) ketone
 (3) acid chloride (4) alkyl chloride
64. The dehydration of 1-butanol gives
 (1) 1-butene as the main product
 (2) 2-butene as the main product
 (3) equal amounts of 1-butene and 2-butene
 (4) 2-methyl propane
65. Glycerol has
 (1) one 1° and one 2° alcoholic groups
 (2) one 1° and two 2° alcoholic groups
 (3) two 1° and one 2° alcoholic groups
 (4) two 2° alcoholic group
66. Reaction of tertiary butyl alcohol with hot Cu at 350°C produces
 (1) butanol
 (2) butanal
 (3) 2-butene
 (4) 2-methyl propene
67. The acidic character of 1° , 2° , 3° alcohols H_2O and $\text{RC}\equiv\text{CH}$ is in the order
 (1) $\text{H}_2\text{O} > 1^\circ > 2^\circ > 3^\circ > \text{RC}\equiv\text{CH}$
 (2) $\text{RC}\equiv\text{CH} > 3^\circ > 2^\circ > 1^\circ > \text{H}_2\text{O}$
 (3) $1^\circ > 2^\circ > 3^\circ > \text{H}_2\text{O} > \text{RC}\equiv\text{CH}$
 (4) $3^\circ > 2^\circ > 1^\circ > \text{H}_2\text{O} > \text{RC}\equiv\text{CH}$
68. Predict the major product

 (1) $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$
 (2) $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
 (3) $\text{I}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$
 (4) no reaction

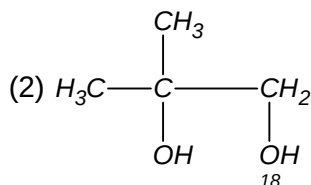
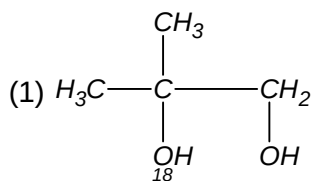
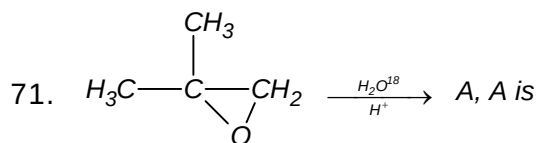
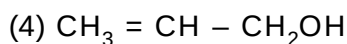
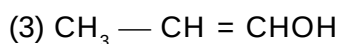
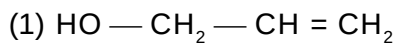
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Alcohol B reacts fastest with Lucas reagent. Hence A and B are



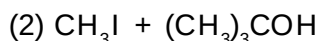
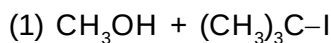
70. Vinyl carbinol is



(3) Both

(4) none

72. Tert-butyl methyl ether on heating with HI of one molar concentration gives



(4) none of these

73. Carbon monoxide acts as a Lewis base because it has

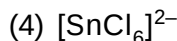
(1) A double bond between C and O atoms

(2) A triple bond between C and O atoms

(3) A lone pair of electrons on the C atom

(4) Two lone pairs of electrons on the O atom

74. Which does not exist ?



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75. Diamond and Graphite have in common

 - (1) Their crystalline structures are identical
 - (2) They are both forms of carbon
 - (3) Both are extremely hard substances
 - (4) The same volume of each substance contains the same number of atoms

76. Which of the following statements is incorrect ?

 - (1) The atoms of carbon in diamond are arranged in a series of interlacing hexagons to form a giant molecule
 - (2) In diamond each atom is joined to four other carbon atoms by covalent linkages
 - (3) Graphite consists of a series of layers each one carbon atom thick in which the carbon atoms are joined in hexagonal rings
 - (4) In graphite there are van der Waals linkages between the adjacent planes

77. Which of the following statements about H_3BO_3 is not correct

 - (1) It is a strong tribasic acid
 - (2) It is prepared by acidifying an aqueous solution of borax
 - (3) It has a layer structure in which planar BO_3 units are joined by hydrogen bonds
 - (4) It does not act as a proton donor but acts as a Lewis acid by accepting hydroxyl ion

78. The type of hybridisation of boron in diborane is

 - (1) sp -hybridisation
 - (2) sp^2 -hybridisation
 - (3) sp^3 -hybridisation
 - (4) sp^3d^2 -hybridisation

79. In the reaction $B_2O_3 + C + Cl_2 \rightarrow A + CO$. The A is

 - (1) BCl_3
 - (2) BCl_2
 - (3) B_2Cl_2
 - (4) CCl_2

80. Identify the statement that is not correct as far as structure of diborane is concerned

 - (1) There are two bridging hydrogen atoms in diborane
 - (2) Each boron atom forms four bonds in diborane
 - (3) The hydrogen atoms are not in the same plane in diborane
 - (4) All B-H bonds in diborane are similar

81. Which of the following is known as inorganic benzene

 - (1) Borazine
 - (2) Boron nitride
 - (3) *p*-dichlorobenzene
 - (4) Phosphonitrilic acid

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- | | |
|--|--|
| <p>82. Carbon and silicon belong to (IV) group. The maximum coordination number of carbon in commonly occurring compounds is 4, whereas that of silicon is 6. This is due to</p> <p>(1) Large size of silicon
 (2) More electropositive nature of silicon
 (3) Availability of low lying d-orbitals in silicon
 (4) Both (1) and (2)</p> <p>83. Carbon suboxide C_3O_2 has</p> <p>(1) Linear structure
 (2) Bent structure
 (3) Trigonal planar structure
 (4) Distorted tetrahedral structure</p> <p>84. Which gas is liberated when Al_4C_3 is hydrolysed</p> <p>(1) CH_4 (2) C_2H_2
 (3) C_2H_6 (4) CO_2</p> <p>85. Which of the following compounds of elements in group IV would you expect to be most ionic in character</p> <p>(1) CCl_4 (2) $SiCl_4$
 (3) $PbCl_2$ (4) $PbCl_4$</p> <p>86. Which of the following compounds of lead is used in match industry</p> <p>(1) PbO
 (2) PbO_2
 (3) $PbCl_2$
 (4) None of these</p> | <p>87. Soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey powder. This transformation is related to</p> <p>(1) A change in the partial pressure of oxygen in the air
 (2) A change in the crystalline structure of tin
 (3) An interaction with nitrogen of the air at very low to temperatures
 (4) An interaction with water vapour contained in the humid air</p> <p>88. Lead pipes are not suitable for drinking water because</p> <p>(1) A layer of lead dioxide is deposited over pipes
 (2) Lead reacts with air to form litharge
 (3) Lead reacts with water containing air to form $Pb(OH)_2$
 (4) Lead forms basic lead carbonate</p> <p>89. PbO_2 is</p> <p>(1) Basic (2) Acidic
 (3) Neutral (4) Amphoteric</p> <p>90. Which of the following does not exist in free form</p> <p>(1) BF_3 (2) BCl_3
 (3) BBr_3 (4) BH_3</p> |
|--|--|

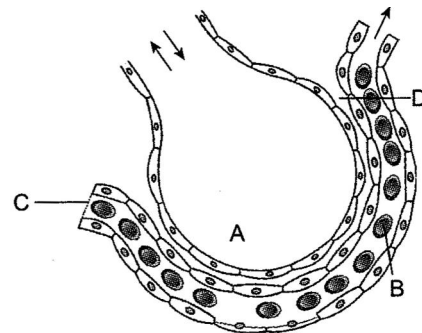
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91. Dead space air is :-
 (1) The amount of air remaining in the alveoli
 (2) The amount of air left behind in lungs at the end of deep expiration
 (3) The amount of air taken in and out
 (4) The air left in the bronchial tree
92. Which Respiratory capacity is correctly matched with its formula & value.

Respiratory Capacities	Formula	Value
(1) Functional Residual Capacity	ERV + TV	2300 ml
(2) Vital Capacity	ERV + TV + IRV	4600 ml
(3) Total Capacity	RV + IRV + ERV	5800 ml
(4) Inspiratory Capacity	TV + IRV	1600 ml

93. In which of the following respiratory tubule dead space volume is not found ?
 (1) Primary bronchi
 (2) Alveolar duct
 (3) Segmental bronchi
 (4) Intra Pulmonary bronchioles
94. Total volume of air, a person can expire after a normal inspiration -
 (1) TV + IRV
 (2) Only TV
 (3) TV + ERV
 (4) TV + FRV + IRV
95. The conducting part of respiratory tract does not help in :-
 (1) Transport the atmospheric air to the alveoli.
 (2) Clear air from foreign particles
 (3) Humidify and bring the air to body temperature
 (4) Diffusion of O_2 and CO_2 between blood and air

96. Total volume of air accommodated in the lungs at the end of a forced inspiration :
 (1) ERV + RV
 (2) TV + ERV
 (3) TV + ERV + IRV
 (4) TV + ERV + IRV + RV
97. Which statements is incorrect ?
 (1) Volume of air remaining in the lungs after a normal expiration is 2300 ml
 (2) Inspiration can occur if the pressure within the lungs is less than atmospheric pressure
 (3) A healthy man can inspire or expire approximately 6000 to 8000 ml of air per minute
 (4) Normal inspiration and expiration both are passive processes.
98. The figure given below shows a small part of human lung where exchange of gases takes place. In which one of the options given below, one part A, B, C or D is correctly identified along with its function,



Options :

- (1) B : Red blood cell—transport of CO_2 mainly
 (2) C : Arterial capillary—passes oxygen to tissues
 (3) A : Alveolar cavity—main site of exchange of respiratory gases
 (4) D : Capillary wall—exchange of O_2 and CO_2 takes place here
99. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of
 (1) Inspiratory reserve Volume
 (2) Tidal Volume
 (3) Expiratory Reserve Volume
 (4) Residual Volume

100. Which of following does not occur during the inspiration
- (1) Contraction of phrenic muscle
 - (2) Increase the volume of thoracic cavity
 - (3) Contraction of internal intercostal muscle
 - (4) Contraction of external intercostal muscle.
101. The maximum volume of air a person can breathe in after a forced expiration is :
- (1) Vital capacity (ERV + TV + IRV)
 - (2) FRC (ERV + RV)
 - (3) EC (TV + ERV)
 - (4) IRV (2500 - 3000 ml)
102. Select an incorrect statement :
- (1) pO_2 of oxygenated blood is 95 mm Hg.
 - (2) CO_2 is 10 times more soluble than O_2 .
 - (3) 70% of CO_2 is transported as bicarbonates
 - (4) 97% of O_2 is transported as oxyhaemoglobin by RBCs.
103. Select a condition which causes the shifting of oxyhaemoglobin dissociation curve to the left :
- (1) High pCO_2 value
 - (2) High temperature
 - (3) Lesser H^+ concentration
 - (4) Decreased pO_2 value
104. What amount of oxygen is delivered to the tissue by 100 ml blood under normal physiological condition ?
- (1) 5 ml
 - (2) 10 ml
 - (3) 15 ml
 - (4) 20 ml
105. Respiratory rhythm centre of medulla is primarily responsible for :
- (1) Regulation of breathing
 - (2) Increased breathing
 - (3) Decreased breathing
 - (4) None of these
106. Which of following have insignificant role in regulation of respiratory rhythm ?
- (1) CO_2
 - (2) H^+ concentration
 - (3) O_2
 - (4) Acidity
107. Which is the occupational pulmonary disease ?
- (1) Pneumonia
 - (2) Emphysema
 - (3) Silicosis
 - (4) Lung cancer
108. The most important muscular structure in respiratory system of human is:-
- (1) EICM
 - (2) IICM
 - (3) Diaphragm
 - (4) Vertebral column
109. Consider the following statements and select the option which includes all the correct ones :-
- (a) Partial pressures of O_2 and CO_2 in Alveolar air are 104 mmHg and 40 mm Hg
 - (b) Partial pressures of O_2 and CO_2 in deoxygenated blood are 40 mmHg and 45 mm Hg respectively
 - (c) Partial pressures of O_2 and CO_2 in deoxygenated blood are 95 mmHg and 40 mm Hg
 - (d) Partial pressures of O_2 and CO_2 in Atmospheric air are 40 mmHg and 46 mm Hg
- Options :-
- (1) Statements a,b and c are correct
 - (2) Statements a, b, c and d are correct
 - (3) Statements a, b are correct
 - (4) Statements c and d are correct
110. What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?
- (1) P_{O_2} lesser, P_{CO_2} higher
 - (2) P_{O_2} higher, P_{CO_2} lesser
 - (3) P_{O_2} higher, P_{CO_2} higher
 - (4) P_{O_2} lesser, P_{CO_2} lesser
111. The Respiratory system is made up of trachea, the lungs, and the?
- (1) Liver
 - (2) Diaphragm
 - (3) Esophagus.
 - (4) Pancreas.

112. When you inhale, your lungs _____.

- (1) Inflate
- (2) Turn purple
- (3) Deflate
- (4) Disappear.

113. The exchange of gases between blood and tissue is referred as _____.:-

- (1) Cellular respiration.
- (2) External respiration.
- (3) Internal respiration.
- (4) All of the above

114. Which one is the cofactor of carbonic anhydrase?

- [illegible]

115. Which one of the following statement is NOT correct regarding trachea?

- (1) It usually lies posterior to the muscular esophagus.
- (2) It splits into the right and left bronchi to supply air to the lungs
- (3) Opening to the trachea is covered by epiglottis.
- (4) Tracheal rings are C-shaped

116. Name the chronic respiratory disorder caused mainly by cigarette smoking.

- (1) Emphysema
- (2) Asthma
- (3) Respiratory acidosis
- (4) Respiratory alkalosis

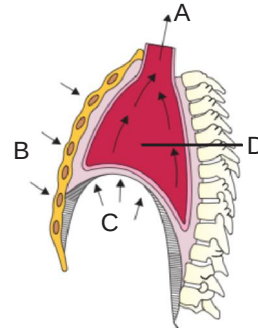
117. Reduction in pH of blood will

- (1) Reduce the rate of heart beat
- (2) Reduce the blood supply to the brain
- (3) Decrease the affinity of haemoglobin with oxygen
- (4) Release bicarbonate ions by the liver

118. Carbonic anhydrase is mostly active in

- (1) RBC
- (2) WBC
- (3) Blood plasma
- (4) Platelets

119. Following diagram depicts the mechanism of breathing. In which of the following options, all the parts A, B, C and D are correctly labelled?



- (1) A-Air expelled from lungs
B-Ribs and sternum raised
C-Diaphragm relaxed
D-Volume of thorax decreased
- (2) A-Air expelled from lungs
B-Ribs and sternum raised
C-Diaphragm contracted
D-Volume of thorax decreased
- (3) A-Air expelled from lungs
B-Ribs and sternum lowered
C-Diaphragm relaxed and arched upwards
D-Volume of thorax decreased
- (4) A-Air entering into lungs
B-Ribs and sternum raised
C-Diaphragm contracted
D-Volume of thorax increased

120. Select the correct combination of the animal and its respiratory organ :-

Animal		Respiratory Structure	
(A)	Prawn	(i)	Dermal branchiae
(B)	Earthworm	(ii)	Tracheae
(C)	Cockroach	(iii)	Moist skin
(D)	Scorpion	(iv)	Gills
(E)	Starfish	(v)	Book lungs

- (1) A-(v), B-(i), C-(ii), D-(iv), E-(iii)
 (2) A-(iv), B-(iii), C-(ii), D-(v), E-(i)
 (3) A-(iv), B-(ii), C-(iii), D-(i), E-(v)
 (4) A-(ii), B-(iii), C-(iv), D-(v), E-(i)

121. A person suffers from punctures in his chest cavity in an accident, without any damage to the lungs. Its effect could be :-
 (1) No change in breathing rate
 (2) Reduced breathing rate
 (3) First increases then reduced breathing rate
 (4) Cessation of breathing
122. Hamburger's phenomenon is also called
 (1) Bicarbonate shift
 (2) Chloride shift
 (3) Hydrogen shift
 (4) Sodium shift
123. During inspiration, air moves into lungs because :-
 (1) Pressure within the lungs and atmospheric air is same
 (2) Pressure within lungs is more than atmospheric pressure
 (3) Pressure within lungs is less than the atmospheric pressure
 (4) Passive flow of air takes air inside
124. Which of the following statements is not true :-
 (1) The partial pressure of O_2 in deoxygenated blood is 40 mm Hg.
 (2) The PO_2 in oxygenated blood is 95 mm Hg.
 (3) The PO_2 in alveolar air is 104 mm Hg.
 (4) The PCO_2 in deoxygenated is 95 mm Hg.
125. Match the items in column-I with column-II and choose the correct option :-

Column-I		Column-II	
A.	TV	I.	2500 to 3000 ml
B.	IRV	II.	1000 ml
C.	ERV	III.	500 ml
D.	RV	IV.	4800 ml
E.	VC	V.	1200 ml

- | A | B | C | D | E |
|---------|-----|----|---|----|
| (1) III | IV | II | I | V |
| (2) III | I | II | V | IV |
| (3) III | I | IV | V | IV |
| (4) IV | III | II | I | V |

126. Read the following four statements (A-D)
 (i) Chest expand because air enters into the lungs
 (ii) Air enters into the lungs because chest expands
 (iii) The muscles of the diaphragm contracts because air enters in the lungs
 (iv) A healthy human breathes 12-16 times/minute
 How many of the statement are wrong?
 (1) Four (2) Three
 (3) Two (4) One
127. Identify false statement :-
 (1) Dead space air is 150 ml
 (2) Air filled in bronchial tree is the dead space air
 (3) Bronchial tree collapses during expiration
 (4) Respiratory tree is inflated during inspiration
128. Myoglobin binds _____ molecule of oxygen.
 (1) 4 (2) 1 (3) 3 (4) 2
129. In methaemoglobin, iron is present in the form of
 (1) Ferrous ions
 (2) Ferric ions
 (3) Ferrous sulphate
 (4) Carboxyhaemoglobin
130. Which of the following statement correctly defines Bohr's effect?
 (1) Rise in P_{50} of haemoglobin with increase in CO_2 concentration
 (2) Fall in oxygen affinity of haemoglobin with increase in CO_2 concentration
 (3) Shift of oxygen dissociation curve of haemoglobin to right with increase in CO_2 concentration
 (4) All of these
131. Volume of oxygen present in one litre of deoxygenated blood under normal condition is
 (1) 50 ml (2) 150 ml
 (3) 200 ml (4) 25 ml

132. In comparison to adult haemoglobin, the oxygen-haemoglobin dissociation curve of foetal haemoglobin is situated towards
- (1) Left side
 - (2) Right side
 - (3) Either right or left
 - (4) None
133. Yawning occurs due to
- (1) Excess of CO_2
 - (2) Deficiency of oxygen
 - (3) Excess of oxygen
 - (4) Deficiency of CO_2
134. Chemoreceptors located in medulla respond to :-
- (1) High level of O_2 in blood
 - (2) Low level of O_2 in blood
 - (3) High level of CO_2 in blood
 - (4) Low level of hydrogen ions in blood
135. Which statement is wrong about partial pressure of oxygen.
- (1) P_{O_2} more in atmospheric air than alveoli
 - (2) P_{O_2} more in alveoli than tissue
 - (3) P_{O_2} more in oxygenated blood than deoxygenated blood
 - (4) P_{O_2} more in deoxygenated blood than oxygenated blood
136. Which point is incorrect about transport of gases?
- (1) Nearly 20-25 percent of CO_2 is transported by RBC
 - (2) About 7 percent of CO_2 is carried in a dissolved state through plasma.
 - (3) About 97 percent of CO_2 is transported by plasma of blood.
 - (4) About 3 percent of O_2 is carried in a dissolved state through the plasma.
137. When we estimate the total volume of air a person can inspire forcefully after normal expiration by spirometer, it indicates –
- (1) TV + ERV
 - (2) TV + IRV
 - (3) ERV + RV
 - (4) VC
138. Muscles involved in effortless breathing as well as breathing with efforts are respectively –
- (1) Internal intercostal muscles, external intercostal muscles
 - (2) External intercostal muscles and Abdominal muscles.
 - (3) Internal intercostal muscles and Abdominal muscles
 - (4) External intercostal muscles and Internal intercostal muscles & abdominal muscles.
139. Conduction zone in the air way is not involved in
- (1) Keeping out harmful microbes
 - (2) Bringing air out of the body
 - (3) Bringing air at body temperature
 - (4) Exchange of gases.
140. Haemoglobin has maximum affinity with–
- (1) CO_2
 - (2) O_2
 - (3) CO
 - (4) NO_2
141. Contraction of external intercostal muscles causes –
- (1) Elevation of ribs and sternum causing increase in volume of thoracic cavity in the dorso-ventral axis.
 - (2) Elevation of the ribs and the sternum causing increase is the volume of thoracic cavity in anterior-posterior axis.
 - (3) Depression of the ribs and the sternum causing increase in volume of thoracic cavity in dorso-ventral axis.
 - (4) Depression of the ribs and the sternum causing increase in the volume of thoracic cavity in anterior-posterior axis.
142. In fever breathing rate –
- (1) Increase
 - (2) Decrease
 - (3) Stop
 - (4) None

143. When P_{50} value of Hb is increased, the union of O_2 with Hb will
 (1) remain unchanged
 (2) increase
 (3) decrease
 (4) Not definite
144. Amount of oxygen present in one gram of haemoglobin is-
 (1) 20 ml (2) 1.34 ml
 (3) 13.4 ml (4) 134 ml
145. Which of the following has no specialised respiratory structures ?
 (1) Crab (2) Earthworm
 (3) Salmon (4) Ant
146. Which statement define functional residual capacity (FRC)
 (1) Total volume of air a person can expire after normal inspiration
 (2) Volume of air that will remain in lungs after normal expiration
 (3) Total volume of air in lungs at the end of forced inspiration
 (4) Volume of air a person can inspire after a normal expiration
147. Arrange the following in increasing order
 (A) Tidal volume
 (B) Residual volume
 (C) Inspiratory reserve volume
 (D) Vital capacity
 (1) $A < B < C < D$ (2) $A < D < C < B$
 (3) $A < C < B < D$ (4) $A < D < B < C$
148. Mark the correct sequence during diffusion of CO_2 from blood to alveoli.
 (a) Alveolar epithelial basement membrane
 (b) Capillary endothelial membrane
 (c) Thin interstitial space
 (d) Alveolar epithelium
 (e) Capillary basement membrane
 (1) d, a, c, e, b (2) e, d, c, b, a
 (3) b, e, c, a, d (4) a, b, c, d, e
149. Every 100 ml of deoxygenated blood deliversml of CO_2 to the alveoli :
 (1) 5 ml (2) 20 ml
 (3) 4 ml (4) 10 ml
150. Which of the following pigment has maximum affinity with O_2 :
 (1) Myoglobin
 (2) Foetal hemoglobin
 (3) Adult hemoglobin
 (4) Haemocyanin
151. Occupational pulmonary disease is :
 (1) Byssinosis (2) Silicosis
 (3) Asbestosis (4) All of the above
152. Conducting part of the respiratory passage is –
 (1) Nostrils to terminal bronchioles
 (2) Nostrils to alveoli
 (3) Nostrils to primary bronchi
 (4) Trachea to alveoli
153. Chloride shift in alveoli takes place from
 (1) Plasma to RBC
 (2) RBC to plasma
 (3) In both directions
 (4) It depends on WBCs
154. Cells of malignant tumours sloughed from such tumours, reach distant sites through blood and wherever they get lodged in the body, they start a new tumor there. This property is called as -
 (1) Contact inhibition
 (2) Metastasis
 (3) Metamorphosis
 (4) Proto- oncogenesis
155. A piece of suspected tissue cut into sections is stained and examined under microscope (histopathological studies) by a pathologist, This process is known as -
 (1) CT-scanning
 (2) Biopsy
 (3) Immunomodulation
 (4) MRI
156. Cancer causing viruses are called oncogenic viruses have genes called viral oncogenes. Which of the following is (are) oncogenic virus(s) ?
 (1) Human papilloma virus
 (2) Kaposi sarcoma associated virus
 (3) Hepatitis-B virus
 (4) All of these

157. Contact inhibition is property of -
 (1) Malignant tumors
 (2) Normal cells
 (3) Cancer cells
 (4) Both (1) and (3)
158. α - interferons can be used in cancer treatment as they act as -
 (1) Mitotic inducer
 (2) Meiotic inhibitors
 (3) viral multiplier
 (4) Immuno modifiers
159. Which of the following gene is found in normal cells, which when activated under certain conditions could lead to oncogenic transformation of the cells ?
 (1) c-oncogene (2) proto oncogene
 (3) Oncogene (4) Both (1) and (2)
160. Diseases can be broadly grouped inA.... andB....
 (1) If 'A' refers to infectious, cancer is its example.
 (2) If 'B' refers to non- infections, AIDS is its example
 (3) If 'A' refers to non-infectious, AIDS and cancer both are its example.
 (4) If 'B' refers to infectious, cancer is not its example.
161. If few disease of human beings are grouped in to following catagories -
 (A) Bacterial disease
 (B) Viral diseases
 (C) Protozoanal diseases
 (D) Helminthic diseases
 Select the groups in which all diseases belong to same group.
 (I) Typhoid, Pneumonia, T.B.
 (II) AIDS, Ascariasis, Rabies
 (III) Polio, AIDS, Typhoid
 (IV) Common cold, Polio, AIDS, Rabies, Hepatitis-B, Ring worm
 (1) I and II (2) II and III
 (3) II and IV (4) Only I
162. The time lag between the infection and appearance of AIDS symptoms is usually
 (1) 5 months – 10 months
 (2) 1 months – 5 months
 (3) 10 years – 20 years
 (4) 5 years – 10 years
163. Spreading of HIV infection can be prevented by -
 (1) Ensuring the use of disposable needles
 (2) Free distribution of condoms
 (3) Advocating safe sex
 (4) All of these
164. Which is incorrect about typhoid fever :
 (a) the pathogen enter in the small intestine through water and food and causes 'intestinal perforation' in the severe condition of the disease
 (b) sustained high grade fever (41°C to 45°C)
 (c) loss of appetite, constipation and generalise weakness are common symptoms of this disease
 (d) salmonella typhi is a kind of bacteria which can spread to the other organs through blood
 (e) the pathogen also infect the large intestine and causes diarrhoea
 (1) a, b, c, d correct while e is incorrect
 (2) a, c, d are correct while b, e are incorrect
 (3) a, b, d are correct while c, e are incorrect
 (4) c, d, e are correct while a, b are incorrect
165. Choose incorrect about Pneumonia disease :
 (1) In pneumonia air filled sacs are infected
 (2) In severe condition of pneumonia cyanosis occurs in the lips, finger nails, ear lobes etc.
 (3) Healthy person acquires this disease by aerosol sharing of glasses, utensils of infected person
 (4) Heamophilus influenzae also damage or infect nasal and pharyngeal mucosa

166. Choose wrong statement about "Amoebic dysentery":
- (1) Entamoeba histolytica is a protozoan parasite which infect large intestine
 - (2) Houseflies are carriers and they transmit the parasite from faeces of infected person to food and water and contaminate them
 - (3) Constipation, Abdominal cramps and stool with excess mucous and blood clots are common symptoms
 - (4) Eggs of Entamoeba are present in the faeces of infected person
167. Choose a correct statement regarding to Ascariasis :
- (1) In this disease person acquires infection through contaminated water, vegetables, fruits etc.
 - (2) It is an intestinal parasite causes internal bleeding due to damage of intestinal mucosa
 - (3) Abdominal pain, fever, anaemia & constipation are common symptoms
 - (4) All are correct
168. A disease which is a slow developing in nature and cause chronic inflammation of the lymphatic vessels, specially in the lower limb vessels, it is :
- (1) Filariasis
 - (2) Ascariasis
 - (3) Amoebiasis
 - (4) Pneumonitis
169. A person having dry skin, scaly lesions on various parts of body such as skin, nails and scalp and intense itching is present. This disease may be due to infection of :
- (1) Fungi
 - (2) Bacteria
 - (3) Virus
 - (4) All of these
170. Aedes mosquito act as vector in :
- (1) Dengue
 - (2) Chickungunia
 - (3) Ascariasis
 - (4) (1) & (2) both
171. A person who is detected positive to ELISA test. The next best test to confirm the disease is :
- (1) Western blot
 - (2) Southern blot
 - (3) Northern blot
 - (4) Eastern blot
172. A person suffering from a disease caused by Plasmodium experiences recurring chill and fever at a time when :
- (1) Sporozoite released from RBC are being rapidly killed and broken down inside spleen
 - (2) Trophozoites reach at maximum growth and release certain toxins
 - (3) Parasite after its rapid multiplication inside the RBC's ruptures them to release merozoites and toxins
 - (4) Microgametocytes and megagametocytes are being destroyed by WBCs
173. Rhino-viruses represent one such group of viruses which cause one of the most infections human ailments the and they infect and but not the
- (1) common cold, nose, respiratory passage, lungs
 - (2) common cold, trachea, bronchi, lungs
 - (3) trachea, bronchi, bronchiole, lungs
 - (4) lungs, nose, respiratory passage, common cold
174. Infective stage of Plasmodium is called :
- (1) Trophozoite
 - (2) Sporozoite
 - (3) Macromerozoites
 - (4) All of these
175. Consider the given diseases : Typhoid, pneumonia, dysentery, plague, common cold, diphtheria, malaria and ringworm.
How many of these diseases are caused by bacterial pathogens :
- (1) Four
 - (2) Five
 - (3) Six
 - (4) Seven
176. Which of following is not a symptom of typhoid :
- (1) Sustained high fever
 - (2) Spasmodic contraction in limbs
 - (3) Intestinal perforation in severe cases
 - (4) Gastric pain and loss of appetite

- | | |
|---|---|
| <p>177. Pneumonia is a bacterial disease which is caused by:</p> <ul style="list-style-type: none">(1) <i>Streptococcus pneumoniae</i>(2) <i>Haemophilus influenzae</i>(3) <i>Clostridium tetani</i>(4) Both (1) & (2) | <p>179. Select the pathogen of fungal disease :</p> <ul style="list-style-type: none">(1) Trichophyton(2) Epidermophyton(3) Microsporum(4) All of these |
| <p>178. Which of following species of plasmodium causes malignant malaria (fatal form of malaria) :</p> <ul style="list-style-type: none">(1) <i>P. vivax</i>(2) <i>P. ovale</i>(3) <i>P. malariae</i>(4) <i>P. falciparum</i> | <p>180. Asexual reproduction of malarial parasite occurs in :</p> <ul style="list-style-type: none">(1) Human RBC(2) Stomach/crop of mosquito(3) Human hepatocytes(4) Both (1) and (3) |



Space For Rough Work
